

CONFIDENCE INTERVALS FOR REGRESSION PARAMETERS BASED ON TRIMMED MEANS

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Key words: Bootstrap; confidence intervals; regression;
robust; simulation; swindle; trimmed means.

Summary: In this paper we use the regression trimmed means, introduced by Koenker and Bassett (1978), to construct confidence intervals for the regression parameters. In this, two approaches are followed. The first approach is similar to that of Gross (1977) where he used the bisquare M-estimators to construct confidence intervals for the regression parameters. The second approach is based on the bootstrap introduced by Efron (1979). We show that both these approaches fare reasonably well in terms of robustness of efficiency.

1. INTRODUCTION

Consider the linear regression model

$$(1.1) \quad \underline{Y} = X\underline{\beta} + \underline{e}$$

with $\underline{Y}$ an $n \times 1$ vector containing the dependent variables, X an $n \times p$ fixed design matrix with first column the vector $\underline{1}$, $\underline{\beta}$ a $p \times 1$ vector

STMA 1985 SUBJECT CLASSIFICATION: 07:150 (04:060)

of unknown regression parameters and $\underline{e}$ an $n \times 1$ vector of independent identically distributed errors with unknown distribution F which is assumed symmetric around 0. Note that traditionally F is taken to be the normal or Gaussian distribution in which case the least squares estimator is the "optimal" estimator.

In the last few years a number of alternative estimators for $\underline{\beta}$ have been proposed that are more robust to deviations from normality than the usual least squares. Of these, perhaps the best known are the M-estimators which are direct extensions of the M-estimators for location (see e.g. Huber (1981)). These estimators have received a considerable amount of attention in the literature and were used *inter alia* to construct robust confidence intervals for $\underline{\beta}$. The latter was done by Gross (1977) where he studied a family of bisquare M-estimators. In this paper we consider regression trimmed means introduced by Koenker and Bassett (1978), as alternatives to least squares to obtain robust confidence intervals for $\underline{\beta}$. In doing this two approaches are followed. The first follows along the lines of the study by Gross (1977) while the second uses the bootstrap introduced by Efron (1979).

As in Gross (1977), we consider linear combinations of $\underline{\beta}$. Thus if $\underline{c}$ is a known vector and $\underline{\beta}_E$ an estimator for $\underline{\beta}$, then a $100(1-\gamma)\%$ confidence interval for $\underline{\ell} = \underline{c}'\underline{\beta}$ may be formed as

$$(1.2) \quad (\underline{c}'\underline{\beta}_E + t_{\gamma/2} D_E, \underline{c}'\underline{\beta}_E + t_{1-\gamma/2} D_E)$$

with D_E a standard error of $\underline{c}'\underline{\beta}_E$ and $t_{\gamma/2}(t_{1-\gamma/2})$ the (estimated) lower (upper) $\gamma/2$ -th quantile of the distribution of $\underline{c}'(\underline{\beta}_E - \underline{\beta})/D_E$. Note the analogy to the location case and in particular the confidence interval for the sample mean. In constructing the intervals (1.2) it is important to find suitable scale estimators for $\underline{\beta}_E$ as well as good estimates for $t_{\gamma/2}$. It is well known that the confidence intervals (1.2) that are obtained when using the least squares estimator tend to be conservative when sampling from a distribution with heavier tails than the normal. That is, the coverage probability is higher than it should be and thus the

interval is not very efficient in terms of expected length. What is desirable from the point of view of robustness is to find a pair $c'\beta_E$ and D_E which give relatively short confidence intervals and a constant t -value over a variety of distributions. This is often referred to as achieving robustness of efficiency and validity.

In the second approach, the bootstrap methodology is used to obtain estimates of the quantiles in (1.2). Monte Carlo methods are used to evaluate and compare these intervals.

The paper is organized as follows. In the next section the regression trimmed mean is defined. In section 3 several scale estimators are proposed and in section 4 the two approaches that are followed in the construction of the confidence intervals are discussed in more detail. The Monte Carlo design is discussed in section 5 and the Monte Carlo results in section 6. The last section contains some concluding remarks.

2. THE REGRESSION TRIMMED MEAN

Unlike the generalization of the M-estimators from the location to the regression case, the generalization of the trimmed mean is not straightforward. A way of generalizing the trimmed mean was provided by Koenker and Bassett (1978) by noting that the sample quantiles in the location case may be obtained as the solution to the minimization problem

$$(2.1) \quad \min_{\mu \in R} \sum_{i=1}^n \rho_{\theta}(Y_i - \mu)$$

with

$$(2.2) \quad \rho_{\theta}(u) = u\psi_{\theta}(u)$$

$$(2.3) \quad \psi_{\theta}(u) = \theta - I(u < 0)$$

and μ the location parameter. Here $Y_1, \dots, Y_n$ denotes a random sample from a distribution F . In this way the quantiles generalize in a straight-

forward way to the regression case. Koenker and Bassett (1978) defined the θ -th regression quantile $\underline{\beta}_{RQ}(\theta)$ as any solution to the minimization problem

$$(2.4) \quad \min_{\underline{\beta} \in R^p} \sum_{i=1}^n \rho_{\theta}(Y_i - \underline{x}_i' \underline{\beta}).$$

Note that $\underline{\beta}_{RQ}(\theta)$ is an M-estimator and that $\underline{x}_i'$ is the i -th row of X . Having defined regression quantiles, the 100 α % regression trimmed mean $\underline{\beta}_{KB}(\alpha)$ (KB for Koenker-Bassett) may now be defined as any solution to the minimization problem

$$(2.5) \quad \min_{\underline{\beta} \in R^p} \sum_{i=1}^n b_i(\alpha)(Y_i - \underline{x}_i' \underline{\beta})^2$$

with

$$(2.6) \quad b_i(\alpha) = \begin{cases} 1 & \underline{x}_i' \underline{\beta}_{RQ}(\alpha) \leq Y_i \leq \underline{x}_i' \underline{\beta}_{RQ}(1-\alpha) \\ 0 & \text{otherwise.} \end{cases}$$

With $B_{\alpha} = \text{diag}(b_i(\alpha))$, this results in

$$(2.7) \quad \underline{\beta}_{KB}(\alpha) = (X' B_{\alpha} X)^{-1} X' B_{\alpha} Y$$

which is a weighted least squares estimator (with weights depending on the data). Note that the 100 α % regression trimmed mean is the least squares estimator of the observations that remain after putting aside observations $(Y_j, \underline{x}_j)$ having

$$Y_j - \underline{x}_j' \underline{\beta}_{RQ}(\alpha) < 0$$

or

$$Y_j - \underline{x}_j' \underline{\beta}_{RQ}(1-\alpha) > 0.$$

From Rupert and Carroll (1980), theorem 3, the asymptotic distribution of the regression trimmed mean follows as

$$(2.8) \quad \sqrt{n} (\underline{\beta}_{KB}(\alpha) - \underline{\beta}) \xrightarrow{D} N(0, \sigma^2(\alpha, F) Q^{-1})$$

with $\sigma^2(\alpha, F)/n$ the asymptotic variance of the trimmed mean in the location case and $Q = \lim_{n \rightarrow \infty} X'X/n$. Note also the close, $O_p(n^{-\frac{3}{2}})$, relationship of

$\underline{\beta}_{KB}$ with a suitably defined Huber M-estimator (see, e.g., Jurečková (1983)).

By, for example de Wet and Venter (1974), with $\xi_1 = F^{-1}(\alpha)$ and $\xi_2 = F^{-1}(1-\alpha)$.

$$(2.9) \quad \sigma^2(\alpha, F) = (1-2\alpha)^{-2} \left\{ \int_{\xi_1}^{\xi_2} x^2 dF(x) + 2\alpha(F^{-1}(\alpha))^2 \right\}.$$

Note that because of the symmetry of F around 0, $F^{-1}(\alpha) = -F^{-1}(1-\alpha)$.

An obvious estimator for $\underline{c}'\underline{\beta}$ is $\underline{c}'\underline{\beta}_{KB}(\alpha)$. This estimator has asymptotic variance

$$(2.10) \quad n^{-1} \sigma^2(\alpha, F) \underline{c}' Q^{-1} \underline{c}.$$

In order to construct confidence intervals for $\underline{c}'\underline{\beta}$ based on $\underline{c}'\underline{\beta}_{KB}(\alpha)$,

we have to estimate this and thus need an estimator for $\sigma^2(\alpha, F)$. Several such estimators are given in the next section.

3. SCALE ESTIMATORS

The asymptotic variance (2.10) of $\underline{\ell}_{KB}(\alpha) = \underline{c}'\underline{\beta}_{KB}(\alpha)$ can be estimated by

$$(3.1) \quad D^2(\alpha) = n^{-1} S^2(\alpha) \underline{c}' (X'X/n)^{-1} \underline{c}$$

with $S^2(\alpha)$ some estimator for $\sigma^2(\alpha, F)$. Since $\sigma^2(\alpha, F)$ is a Winsorized variance of F , a natural estimator for this is a Winsorized variance based on the (estimated) residuals. To be more precise, let

$$r_i = r_i(\alpha) = y_i - \underline{x}_i' \underline{\beta}_{KB}(\alpha) \quad i=1, 2, \dots, n,$$

be the i -th residual and $r_{(1)} < r_{(2)} < \dots < r_{(n)}$ the ordered values. Our first estimator for $\sigma^2(\alpha, F)$ is given by

$$(3.2) \quad S_1^2(\alpha) = (1-2\alpha)^{-2} \left\{ (n-p)^{-1} \sum_{i=[n\alpha]+1}^{n-[n\alpha]} r_{(i)}^2 + \alpha(r_{([n\alpha]+1)}^2 + r_{(n-[n\alpha])}^2) \right\}$$

with $[x]$ denoting the largest integer contained in x . Consistency of $S_1^2(\alpha)$ follows easily using results of Bickel (1973), remark 3, p. 604.

Another estimator was suggested by Ruppert and Carroll (1980) and modified by Koenker (1983) to make it invariant with respect to a shift in the design. This is given by

$$(3.3) \quad S_2^2(\alpha) = (1-2\alpha)^{-2} \left\{ (n-p)^{-1} \sum_{i=1}^n b_i(\alpha) r_i^2 + \alpha(c_1^2(\alpha) + c_2^2(\alpha)) \right\}$$

where

$$c_1(\alpha) = \bar{x}'(\beta_{RQ}(\alpha) - \beta_{KB}(\alpha))$$

and

$$c_2(\alpha) = \bar{x}'(\beta_{RQ}(1-\alpha) - \beta_{KB}(\alpha)).$$

Here $\bar{x}'$ denotes a row vector consisting of the means of the columns of X . Consistency of $S_2^2(\alpha)$ is shown in Ruppert and Carroll (1980).

REMARKS

- (i) Several variants of $S_1^2(\alpha)$ and $S_2^2(\alpha)$ have been tried but found not to perform better than $S_1^2(\alpha)$ and $S_2^2(\alpha)$ and were thus not considered further. For details of this, see de Jongh (1985).
- (ii) Note that the above estimators are both shift (regression) invariant and scale equivariant. By this is meant the following. Suppose $S^2 = S^2(Y, X)$ is a scale estimator. Then S^2 is shift or regressor

invariant if

$$S^2(\underline{Y} + \underline{\lambda}, \underline{X}) = S^2(\underline{Y}, \underline{X})$$

for any constant p -vector $\underline{\lambda}$, and scale equivariant if

$$S^2(c\underline{Y}, \underline{X}) = c^2 S^2(\underline{Y}, \underline{X})$$

for any constant $c > 0$.

That $S_1^2(\alpha)$ and $S_2^2(\alpha)$ have these properties follows immediately from their definitions and the fact that the regression trimmed means are shift and scale equivariant (the latter follows easily from theorem 3.2 of Koenker and Bassett (1978)).

- (iii) Some remarks on the comparison of $S_1^2(\alpha)$ and $S_2^2(\alpha)$ are given in section 6.

6. CONSTRUCTION OF CONFIDENCE INTERVALS

In this section the methods used in the construction of the confidence intervals for $\ell = c'\underline{\beta}$ are discussed in more detail. Recall that the intervals are of the form (in the symmetric case)

$$(4.1) \quad (\ell_{KB} - t^*D, \ell_{KB} + t^*D)$$

with $\ell_{KB} = \ell_{KB}(\alpha) = c'\underline{\beta}_{KB}(\alpha)$, $D = D(\alpha)$ a standard error of ℓ_{KB} and t^* a constant chosen so that the interval (4.1) has a probability $1-\gamma$ of covering ℓ . Without loss of generality we take $\underline{\beta} = \underline{0}$ and thus $\ell = 0$. Both the approaches will be concerned with estimating t^* and evaluating the intervals with respect to their coverage probabilities and expected lengths (ECIL). These approaches are discussed in the next two sub-sections and will be referred to as methods GROSS and BOOT respectively.

4.1 Method GROSS

This approach follows the lines of Gross (1977) which was carried out for the family of bisquare M-estimators. His main objective was to find a suitable (compromise) t^* value (for given sample size and scale estimator) achieving robustness of validity and efficiency. A secondary objective was to find a suitable degree-of-freedom correction so that available t-tables could be used to give a t^* value for a particular sample size. In this Gross used the so-called Monte Carlo swindle to estimate t^* values and ECIL's for the normal and slash distributions over a number of design matrices and linear combinations. We now briefly describe how the swindle can be used to estimate the above quantities.

The swindle is applicable when the errors e_i can be generated as the ratios Z_i/V_i where the $\{Z_i\}$ are i.i.d. from a normal distribution and the $\{V_i\}$ are also i.i.d. with support a subset of $(0, \infty)$ and independent of the $\{Z_i\}$. For example, to generate values from a slash distribution, the $\{V_i\}$ are from a uniform distribution on $(0, 1)$.

Now, given $\{V_i\}$, the best linear estimator for ℓ is $\hat{\ell} = \underline{c}' \hat{\underline{\beta}}$ with

$$(4.2) \quad \hat{\underline{\beta}} = (X'D_V X)^{-1} X'D_V \underline{Y}$$

where $D_V = \text{Diag}(V_i)$. Let

$$(4.3) \quad S^2 = (\underline{Y} - X\hat{\underline{\beta}})' D_V (\underline{Y} - X\hat{\underline{\beta}}),$$

then it is easy to verify that

$$\hat{\ell} | \underline{V} \sim N(\ell, \sigma_c^2) \quad \text{with} \quad \sigma_c^2 = \underline{c}' (X'D_V X)^{-1} \underline{c}$$

and

$$S^2 | \underline{V} \sim \chi^2_{n-p}.$$

Also, note that $\hat{\ell}$ and $\underline{r} = \underline{Y} - X\hat{\underline{\beta}}$ are conditionally independent. Now, using this, the shift equivariance of the trimmed mean and the shift in-

variance of the scale estimators it follows that

$$(4.4) \quad P(\ell_{KB} - kD > 0) = E[1 - \phi((kD - \ell_{KB} + \hat{\ell})/\sigma_c)]$$

and by symmetry that

$$(4.5) \quad P(\ell_{KB} - kD > 0) = E[\phi((kD - \ell_{KB} + \hat{\ell})/\sigma_c)]$$

For appropriate choices of k , the quantities on the right-hand side (RHS) of (4.4) and (4.5) can be averaged and the average of this over a number of Monte Carlo replicates will yield an estimate of $P(\ell_{KB} - kD > 0)$. The required value of t^* can now be obtained by interpolation so that

$$(4.6) \quad P(\ell_{KB} - t^*D > 0) = \gamma/2$$

This t^* value is then used in estimating the expected confidence interval length (ECIL), with

$$(4.7) \quad \text{ECIL} = 2t^*E(D)$$

Using the shift invariance and scale equivariance of D and conditional independence of $\hat{\beta}$ and r/S , it follows easily that

$$\text{ECIL} = aE(D/S)$$

$$\text{with } a = \sqrt{2}\Gamma[(n-p+1)/2]/\Gamma[(n-p)/2]$$

and $\Gamma(\cdot)$ the gamma function. Again the RHS can be estimated by Monte Carlo in the usual way.

REMARK

Note again that the t^* values and ECIL values will be calculated for combinations of design matrices, linear combinations and scale estimators. From these results and for a specific scale estimator, a compromise t^*

value will be found giving approximately the right coverage and relatively short interval lengths over the different factor combinations considered. The confidence interval used in practice will thus be

$$(4.8) \quad (\ell_{KB}(\alpha) - t_c^* D(\alpha), \ell_{KB}(\alpha) + t_c^* D(\alpha)) ,$$

with t_c^* the compromise t^* value. For details on the latter, refer to section 5.

4.2 Method BOOT

This method uses the bootstrap procedure to construct confidence intervals of the form (4.1). The idea of the bootstrap is relatively new and was first introduced by Efron (1979). In Efron (1982), methods were also discussed for constructing confidence intervals for parameters in the location case by using this idea. According to Efron and Gong (1983) the appropriate theory to justify these methods is still far from clear but, although the work is highly speculative in nature, it has yielded very encouraging results in many examples studied thus far. The methods proposed here are analogues of Efron's bootstrap pivotal and percentile intervals.

Two methods are proposed for constructing $100(1-\gamma)\%$ confidence intervals for ℓ based on the bootstrap. These are now discussed.

Let $(Y_1, x_1'), \dots, (Y_n, x_n')$ be the observed data from the regression model (1.1). Given this data obtain the estimator $\ell_{KB}(\alpha)$ and an appropriate scale estimator $D(\alpha)$. Now, based on the estimator $\beta_{KB}(\alpha)$, form the residuals

$$r_i(\alpha) = Y_i - x_i' \beta_{KB}(\alpha)$$

for $i = 1, 2, \dots, n$. Let $r_1^*(\alpha), \dots, r_n^*(\alpha)$ be a random sample from the empirical distribution function of the $\{r_i(\alpha)\}$, i.e. each $r_i^*(\alpha)$ is drawn independently and with replacement and with equal probability from $r_1(\alpha), \dots, r_n(\alpha)$. Form the pseudo-data (Y_i^*, x_i') as

$$Y_i^* = \underline{x}_i' \underline{\beta}_{KB}(\alpha) + r_i^*(\alpha)$$

with $\{r_i^*(\alpha)\}$ the bootstrap sample. Using this pseudo-data calculate the bootstrap estimators $\ell_{KB}^*(\alpha) = \underline{c}' \underline{\beta}_{KB}^*(\alpha)$ and $D^*(\alpha)$. Here the bootstrap estimators $\underline{\beta}_{KB}^*(\alpha)$ and $D^*(\alpha)$ is the same as $\underline{\beta}_{KB}(\alpha)$ and $D(\alpha)$, except that it is calculated on the pseudo-data instead of the real data. Repeat the resampling procedure B times to obtain B bootstrap estimators

$$\ell_{KB,1}^*(\alpha), \dots, \ell_{KB,B}^*(\alpha).$$

Now, form the upper and lower $\gamma/2$ quantiles of this empirical distribution and call these $h(1-\gamma/2)$ and $h(\gamma/2)$ respectively. An approximate $100(1-\gamma)\%$ confidence interval for $\ell = \underline{c}' \underline{\beta}$ is then

$$(h(\gamma/2), h(1-\gamma/2)).$$

This method will be referred to as BOOT1 and is based on the percentile method of Efron (1982).

For the second method let $g(1-\gamma)$ be the upper γ quantile of the empirical distribution of

$$|\ell_{KB,1}^*(\alpha) - \ell_{KB}(\alpha)|/D_1^*(\alpha), \dots, |\ell_{KB,B}^*(\alpha) - \ell_{KB}(\alpha)|/D_B^*(\alpha)$$

with $D(\alpha)$ either one of the two scale estimators discussed in section 3. An approximate $100(1-\gamma)\%$ confidence interval for $\ell = \underline{c}' \underline{\beta}$ is then

$$(\ell_{KB}(\alpha) - g(1-\gamma)D(\alpha), \ell_{KB}(\alpha) + g(1-\gamma)D(\alpha)).$$

For the two scale estimators in section 3 (based on $S_1^2(\alpha)$ and $S_2^2(\alpha)$), this method will be referred to as BOOT2 and BOOT3 respectively. Note that this method is a variant of Efron's pivotal method.

REMARK

Note that in contrast to method GROSS, the BOOT method does not require a compromise t^* value. On the other hand, it has the disadvantage that it requires much more computing effort to construct a BOOT interval.

5. MONTE CARLO DESIGN

The following five factors are included in the Monte Carlo study:

- error distributions
- design matrices and sample sizes
- linear combinations
- estimators.

The levels that these factors were studied at, are now given.

5.1 Error distributions

The following distributions were considered:

- F1: The normal distribution $N(0,1)$.
- F2: The contaminated normal distribution $CN(0,1;9)$.
- F3: The contaminated normal distribution $CN(0,2;9)$.
- F4: The contaminated normal distribution $CN(0,1;100)$.
- F5: The slash distribution.

Here $CN(\varepsilon, \tau^2)$ denotes a $N(0,1)$ contaminated with a $N(0, \tau^2)$, with contamination probability ε . The slash distribution is generated as Z/U , with Z a $N(0,1)$ random variable and U an independent $U(0,1)$ random variable. This distribution has the heavy tails of the Cauchy distribution but looks more like the normal distribution in the centre. Note that all the distributions are of the type to which the swindle applies.

5.2 Design matrices and sample sizes

Five design matrices were studied for each of three sample sizes $n = 11, 21, 41$. Throughout p is taken as 2, i.e. only one independent variable is considered, since the first column of X consists of ones. Denoting the values of the independent variable by x_i , $i=1,2,\dots,n$, the following design matrices were used

$$\begin{aligned} M1: x_i = & -1 & i = 1, 2, \dots, (n-1)/2 \\ & 0 & i = (n+1)/2 \\ & 1 & i = (n+3)/2, \dots, n. \end{aligned}$$

M2: x_i = the expected value of the i -th order statistic in a sample of size n from the $N(0,1)$ distribution.

$$M3: x_i = i \quad i = 1, 2, \dots, n.$$

$$M4: x_i = i^2 \quad i = 1, 2, \dots, n.$$

$$M5: x_i = 2^i \quad i = 1, 2, \dots, n.$$

These vectors were then transformed to have mean zero and sum of squares equal to n (thus $X'X = nI$). This was done to scale the second column vectors to have the same length as the column of ones.

REMARKS

- (i) Design matrices $M1$, $M2$ and $M3$ are well behaved although $M1$ is rather special in that it consists of two clusters with a single centre point. On the other hand $M4$ and $M5$ are not well behaved since they contain high leverage points.

Note that design $M5$ is rather extreme. In particular, it does not satisfy the regularity assumptions needed to justify (2.8) (condition 3, page 830 in Ruppert and Carroll (1980) does not hold). Also, if the leverage of the i -th data point is taken as the i -th diagonal element of the so-called hat matrix (see, e.g. Cook and

Weisberg (1982)) we find that for M5, $n=21$ most of the leverage is contained in the last 3 data points. The leverage of the first 18 points is almost constant at approximately 0,050 while the last 3 points have leverages of respectively 0,069, 0,191 and 0,763. Thus although M5 is a rather pathological design, it was included to illustrate its adverse effect on robust estimators such as the trimmed means. (See the results in the next section.) Design M4 also contains high leverage points but they are less extreme than M5.

- (ii) Sample sizes of 11, 21, 41 were used as the natural analogues to 10, 20 and 40 for the location case ($p=1$).

5.3 Linear combinations

The linear combinations that were studied are the same as studied by Gross (1977). These are

- L1: $\underline{c}' = (1; 0)$
 L2: $\underline{c}' = (0,95; 0,32)$
 L3: $\underline{c}' = (0,71; 0,71)$
 L4: $\underline{c}' = (0,32; 0,95)$
 L5: $\underline{c}' = (0; 1)$.

L1 and L5 are special cases in the sense that the intercept and slope parameters respectively are studied. The other linear combinations consider different weightings of the two parameters. Note that for the choices of the linear combinations given here, $\underline{c}'\underline{c} = 1$ and thus $\underline{c}'(X'X)^{-1}\underline{c} = \frac{1}{n}$.

5.4 The estimators

Trimming proportions of $\alpha=0; 0,1; 0,2$ and $0,3$ were considered. These estimators will be denoted by $E0(\alpha=0), E1(\alpha=0,1), E2(\alpha=0,2)$ and $E3(\alpha=0,3)$. Note that $E0$ is the least squares estimator.

The Monte Carlo study was conducted in two stages.

First method GROSS was studied. A thousand Monte Carlo replicates ($N=1000$) were used to estimate the t^* and ECIL values for each of the above-mentioned factor combinations. For each estimator and scale estimator a t_C^* value was chosen based on these ECILs and the constancy of the t^* 's over distributions. The t_C^* value was then used to construct a confidence interval of the form (4.8) and again 1000 Monte Carlo replicates were used to estimate the coverage probabilities of these intervals for the different factor levels considered.

Secondly, method BOOT was studied. In this study only 250 bootstrap replicates ($B=250$) and 250 Monte Carlo replicates ($N=250$) were used to estimate respectively the appropriate quantiles and the ECILs and coverage probabilities. The latter quantities were estimated using naive simulation methods.

Throughout γ was taken as 0,05 to yield 95% confidence intervals for ℓ .

All calculations were performed in double precision arithmetic on a VAX/VMS 750 mini-computer. A standard linear congruential random number generator was used.

In the next section the results and a discussion on the results of both studies are given. This is done separately for the two studies since the results are not directly comparable (due to different number of replicates used). However, some concluding remarks regarding the performance of the two methods will be made in the last section.

6. MONTE CARLO RESULTS

6.1 Method GROSS

The simulation results were analyzed by means of a two-way analysis by medians (Tukey (1977)). This was done to represent the effect of linear

combinations (L) and design matrices (M) on t^* and ECIL in tabular form. The results obtained for the two scale estimators and the five distributions for estimator E2 appear in tables 1 (t^* times 100) and 2 (ECIL times 100). All the ECIL and t^* values were multiplied by 100 to simplify interpretation.

Note that the grand effect, L-effect, M-effect and residuals are displayed in each subtable of tables 1 and 2. The original value in each subtable can then be reconstructed by the formula

$$\frac{t^*}{ECIL} = \text{grand effect} + \text{L-effect} + \text{M-effect} + \text{residual}.$$

We will now discuss the results in tables 1 and 2 separately. Only the results for E2 are given since those for the other estimators gave qualitatively similar results. Some comparison across estimators will be done after discussing tables 1 and 2 (by looking at grand effects only).

Results in table 1 (t^* values)

The scale estimator based on $S_2^2(\alpha)$ consistently gives lower t^* values than the one based on $S_1^2(\alpha)$. For $S_2^2(\alpha)$ the grand effects typically are between 191 and 197 while for $S_1^2(\alpha)$ the grand effects are between 250 and 242. For both scale estimators the matrix M5 has an increasingly prominent effect at respectively distributions F3, F4 and F5. This effect is particularly prominent for $S_1^2(\alpha)$. Design matrix M1 also has a noticeable effect for $S_2^2(\alpha)$ over all distributions. L-effects are prominent, except perhaps for the L1-M5-F5 combination for $S_1^2(\alpha)$.

Standard errors were calculated for the t^* values in this table. This was done by repeating the Monte Carlo study twice with 500 replicates each. For the scale estimator based on $S_2^2(\alpha)$ the standard errors were around 23 for all distributions. For the other scale estimator the standard errors ranged from around 21 to 48. Large standard errors were

TABLE 1

t* values (times 100) for E2.

E2												
s ₂						s ₁						
	L1	L2	L3	L4	L5	F1	L1	L2	L3	L4	L5	M-effect
						M-effect						
M1	1	4	0	-9	0	30	0	4	1	-9	0	1
M2	-2	-4	0	0	0	5	-1	-3	0	0	-1	4
M3	-3	-1	0	0	1	0	-3	0	0	0	2	0
M4	0	0	0	1	2	0	0	-1	0	1	1	-1
M5	8	4	0	-7	-6	-4	11	5	0	-10	-8	-6
L-effect	-2	-1	0	5	3	193	-4	-4	0	5	5	231

F2												
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect
M1	3	2	3	-6	0	26	1	1	1	-8	0	-1
M2	3	0	-2	6	5	-1	2	0	0	7	10	0
M3	-1	0	0	-1	6	-4	0	-3	0	-3	8	-2
M4	0	3	-2	0	0	0	-1	2	0	0	0	4
M5	-2	-6	8	11	0	5	-9	-8	15	13	0	13
L-effect	-3	-1	1	1	0	195	-4	-1	0	4	0	230

F3												
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect
M1	8	6	0	-3	-3	19	15	10	-1	-4	-5	-10
M2	0	-5	-1	1	4	0	-1	-7	0	3	3	1
M3	6	0	2	0	-2	-7	12	4	4	0	-5	-6
M4	0	0	0	-2	0	-3	0	0	1	-2	0	0
M5	-15	-13	-1	8	8	16	-26	-18	0	9	8	33
L-effect	-8	-4	1	2	2	197	-13	-7	0	2	5	236

F4												
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect
M1	5	6	-2	-3	0	24	11	8	-1	-2	-5	-7
M2	0	0	-1	1	4	0	5	0	0	6	1	-3
M3	1	2	0	0	0	-3	0	4	-2	-2	0	0
M4	-1	-4	2	-3	3	0	-7	-5	0	0	5	9
M5	-74	-47	14	53	0	72	-135	-65	52	117	-3	170
L-effect	-4	-2	1	3	0	191	-11	-4	0	2	3	237

F5												
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect
M1	13	6	-2	-5	0	13	23	13	0	-5	-4	-17
M2	-1	-7	0	-5	1	0	0	0	0	0	4	0
M3	4	0	1	3	0	-10	6	5	0	4	0	-3
M4	0	0	5	0	0	-2	-10	-2	9	-2	-1	19
M5	-107	-75	-1	73	49	105	-224	-137	0	78	5	281
L-effect	-11	-6	1	5	0	192	-18	-11	0	5	1	242

observed for combinations M5-F4 (around 206) and M5-F5 (around 292).

In order to get a better idea of the effects of the different factors on the t^* values a few graphical representations are now given. Figure 1 contains the t^* values of E2, obtained by using the scale estimator $S_1^2(\alpha)$ and figure 2 those obtained by using scale estimator $S_2^2(\alpha)$. Only the results for the normal (F1) and slash (F5) distributions are given. In both figures the t^* values seem relatively constant over the linear combinations and distributions, except for design matrix M5 at distribution F5. Also matrix M1 shows a distinct effect when using scale estimator $S_2^2(\alpha)$. The remaining design matrices seem to perform very similar in both cases. For this, it is interesting to note that the scale estimator $S_1^2(\alpha)$ yields slightly higher t^* values than does $S_2^2(\alpha)$.

REMARKS

- (i) The above results show a remarkable resemblance to those obtained by Gross (1977) for the M-estimators.
- (ii) Results for the other estimators were similar, although it seemed as if the M5 effect increased with increased trimming.

Results in table 2 (ECIL times 100)

The results show that the two scale estimators give approximately the same confidence interval lengths for E2. Also no L- and M-effects are observed up to and including distribution F3. For distributions F4 and F5 the M5 design shows a large effect. Also note the increasing confidence interval lengths for the heavier tailed distributions.

Standard errors were calculated as before. For both scale estimators the standard errors were typically around 2 for distributions F1 and F4 and around 6 for distribution F5. For the M5-F5 combination the standard errors were around 50 for both scale estimators.

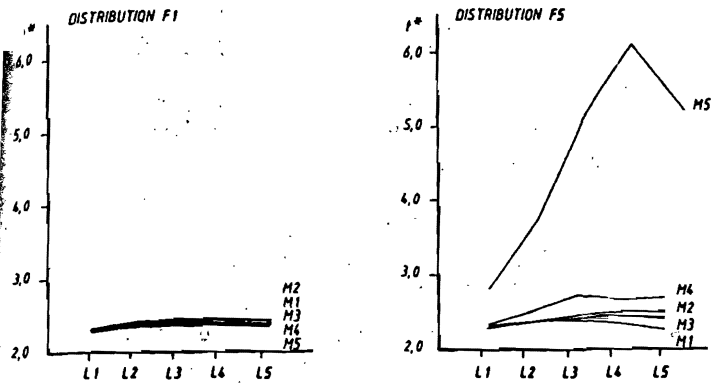


Figure 1. The t^* values of E2 based on scale estimator $S_1^2(\alpha)$.

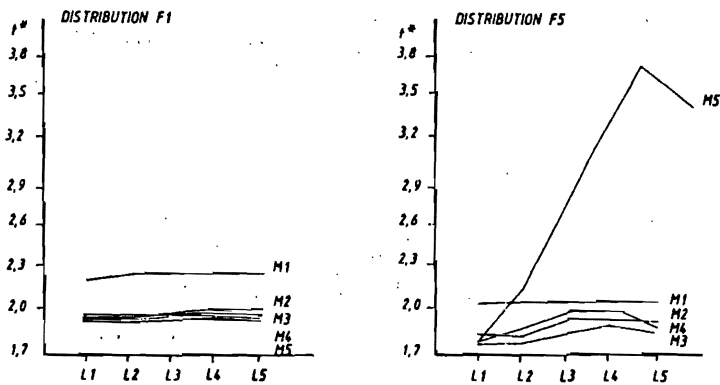


Figure 2. The t^* values of E2 based on scale estimator $S_2^2(\alpha)$.

REMARK

The results for the other estimators were qualitatively similar.

Only the results for estimator E2 were discussed. For more details on the other estimators the interested reader is referred to de Jongh (1985). However, we now give the grand effects of the t^* and ECIL analyses in tables 3 and 4 respectively for all the estimators (and factors).

From table 4 it can be seen that the two scale estimators give almost the same confidence interval lengths whereas, from table 3 the t^* values to achieve these are different for the two scale estimators and in particular $S_2^2(\alpha)$ requires a smaller t^* value. Also note that the t^* values for $S_1^2(\alpha)$ are not as constant as those for $S_2^2(\alpha)$, especially for E1.

Based on the results in table 4 the efficiencies of the four estimators were obtained as

$$\text{efficiency} = \frac{\text{min ECIL of all estimators over distributions}}{\text{ECIL for estimator}} \times 100$$

The resulting values are given in table 5. These indicate E2 to be the "best" regression trimmed mean for a sample size of 21 over nearly all the factors considered in this study.

Compromise t^* value

The t^* values calculated in the previous section can now be used to construct confidence intervals for λ . However, in practice one never knows the shape of the distribution and thus will not know what value of t^* to use. Also one would like to use a t^* value that does well for any design and linear combination. Usually one would like to use the largest t^* value in order to obtain conservative confidence intervals, but this seemed to be too extreme to use especially the t^* values obtained in the M5-F5 cases. Several of the t^* values presented in table 3 were tried out as a possible compromise value. Of these the

TABLE 2

ECIL values (times 100) for E2

E2														
S ₂						S ₁								
						F1								
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect		
M1	1	1	0	-2	0	1	2	1	1	-2	0	0		
M2	-1	-3	0	0	0	2	-1	-3	0	0	-1	3		
M3	0	0	0	0	1	0	-1	0	0	0	0	0		
M4	0	0	0	0	1	0	0	-1	0	1	1	0		
M5	5	2	0	-3	-4	-3	6	2	0	-3	-4	-3		
L-effect	-2	-1	0	2	1	102	-3	-1	0	2	2	103		
F2														
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect		
M1	2	1	0	-2	0	-1	2	2	0	-2	0	-2		
M2	0	0	0	3	1	0	0	0	0	3	2	0		
M3	0	0	1	0	3	-2	1	-1	0	-1	3	-2		
M4	-1	1	0	0	0	1	0	1	0	0	-1	1		
M5	-1	-2	7	9	0	3	-3	-2	8	9	-1	4		
L-effect	-2	-1	0	0	0	114	-3	-1	0	1	1	115		
F3														
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect		
M1	3	4	0	-2	-4	-3	6	5	0	-2	-4	-5		
M2	-1	-2	0	1	2	1	0	-2	0	2	0	1		
M3	3	2	2	0	-4	-2	5	2	2	0	-4	-2		
M4	0	0	2	-2	0	0	0	0	1	-1	0	0		
M5	-12	-8	0	5	1	15	-15	-9	0	6	0	18		
L-effect	-4	-3	0	2	4	127	-6	-4	0	1	4	129		
F4														
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect		
M1	4	2	-1	-2	2	-5	6	2	0	0	0	-4		
M2	1	0	0	2	0	-2	4	0	0	3	-1	-2		
M3	-3	0	0	0	-1	0	0	2	0	0	1	0		
M4	0	-2	0	-2	5	2	-3	-3	0	0	5	7		
M5	-54	-36	11	43	-1	72	-70	-37	30	64	-1	93		
L-effect	-4	-1	0	2	1	124	-7	-1	0	0	1	124		
F5														
	L1	L2	L3	L4	L5	M-effect	L1	L2	L3	L4	L5	M-effect		
M1	22	14	4	-10	-7	-28	23	10	0	-7	-5	-20		
M2	0	0	-3	0	1	0	0	0	0	0	6	0		
M3	9	0	0	2	-3	-3	10	1	0	5	0	-1		
M4	-5	2	11	-10	0	20	-17	-7	12	-4	0	30		
M5	-316	-17	-3	88	2	458	-292	-144	17	101	0	390		
L-effect	-17	-14	3	10	3	289	-20	-10	0	7	0	280		

TABLE 3

Grand effects of t^* -analysis (times 100) for four estimators, two scale estimators and five distributions.

	F1		F2		F3		F4		F5	
	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1
E0	200	199	197	195	196	193	193	188	198	192
E1	183	218	184	230	184	233	184	265	183	271
E2	193	231	195	230	197	236	191	237	192	242
E3	215	256	215	257	218	256	211	255	213	260

TABLE 4

Grand effect of ECIL (times 100) for four trimming proportions, five distributions and two scale estimators.

	F1		F2		F3		F4		F5	
	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1
E0	91	91	118	118	140	141	259	258	2549	2583
E1	96	96	114	115	134	134	169	167	473	434
E2	102	103	114	115	127	129	124	124	289	280
E3	116	117	126	127	138	139	130	132	278	274

TABLE 5

Efficiencies of estimators (min ECIL/ECIL).

	F1		F2		F3		F4		F5	
	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2	S_1
E0	100	100	97	97	91	91	45	48	11	11
E1	95	95	100	100	95	96	73	74	59	63
E2	89	88	100	100	100	100	100	100	96	98
E3	78	78	90	91	92	93	95	94	100	100

average t^* value (grand effect) obtained from the (contaminated) normal distribution F3 when using scale estimator $S_2^2(\alpha)$ seemed to yield the best results overall. The coverage probabilities obtained by using this t_c^* value appear in table 6 for E2 for five design matrices, five distributions and three linear combinations. The coverage probabilities were calculated by using naive Monte Carlo methods.

Almost all the values obtained for designs M2, M3 and M4 are within 1,5% of the 95% coverage goal for all the distributions considered. It is only at the M1 and M5 designs that some problems arise. For instance the coverage probabilities obtained under M1 is a bit on the low side over all linear combinations and distributions studied. The same applies to the M5 design at linear combinations L3 and L5. Note again the alarming values obtained in the M5-F5 cases.

None the less if one would consider distribution F5 and designs M1 and M5 rarities in practice, the results are very encouraging indeed.

Degrees of freedom correction

The above discussion only applies to a sample size of 21. In order to get an idea of the behaviour of the estimators under the same conditions for other sample sizes, the whole study was repeated for $n=11$ and $n=41$. This was done mainly to obtain a suitable formula to adjust the degrees of freedom so that ordinary t-tables may be used to obtain critical values. The attempt was without avail since no discernible pattern emerged from the three studies as far as the proposed degrees of freedom corrections were concerned. It was decided not to pursue this idea further, but rather to use a technique such as the bootstrap to obtain confidence intervals. The results of this are discussed next.

6.2 Method BOOT

As mentioned previously, naive simulation methods and a different number of Monte Carlo replicates ($N=250$) were used compared to method GROSS. The

TABLE 6

Coverage probabilities (times 100) obtained by using estimator E2 and method GROSS based on $S_2^2(\alpha)$ and the t^* value (t^* grand effect) for F3 as compromise.

L1					
	M1	M2	M3	M4	M5
F1	93,5	94,0	94,6	95,5	96,4
F2	92,3	95,0	94,7	96,3	95,2
F3	93,2	96,4	95,5	95,5	95,0
F4	92,6	95,9	96,5	95,5	96,4
F5	93,3	96,2	96,7	96,1	97,4

L3					
	M1	M2	M3	M4	M5
F1	91,3	95,2	96,3	96,0	95,9
F2	93,4	94,7	95,0	95,4	94,9
F3	94,2	95,0	95,0	94,9	93,4
F4	91,6	96,1	95,5	96,2	90,1
F5	94,9	95,6	95,4	95,1	89,8

L5					
	M1	M2	M3	M4	M5
F1	92,9	94,7	94,9	95,0	95,3
F2	91,2	95,0	94,5	94,7	94,5
F3	93,2	95,0	95,8	94,5	92,8
F4	92,9	96,0	95,4	95,4	92,5
F5	94,6	95,5	96,4	96,0	88,1

results are thus not directly comparable to the previous section.

Owing to limitations on computer central processing time the linear combinations were limited to L1, L3 and L5 and the designs to M1, M3 and M5. The ECILs and coverage probabilities for E2 appear in table 7. Estimated "true" confidence interval lengths (TRU) were obtained by calculating the interval $(k(\gamma/2), k(1-\gamma/2))$ where $k(\gamma/2)$ is the $\gamma/2$ quantile of the Monte Carlo distribution of $\ell_{KB}(0,2)$ obtained in 250 replicates.

The confidence interval results in table 7(a) show that the confidence interval lengths obtained for BOOT1 are closest to TRU over nearly all the M and L combinations for distributions F1 and F2. For distributions F4 and F5 methods BOOT2 and BOOT3 are the better ones in this respect, while for F3 the three methods perform similarly. The confidence interval lengths seem to be reasonably constant over designs and linear combinations, for each fixed distribution. As usual, the M5-F5 and M5-F4 cases behave rather differently and all the above conclusions have been made ignoring these cases.

Standard errors were again calculated. Method BOOT1 had the smallest standard errors for F1 ($\pm 1,1$), F2 ($\pm 1,5$) and for F3 (± 2) while method BOOT3 had smallest standard errors for F4 ($\pm 4,5$) and F5 (± 10).

From table 7(b) it is clear that the coverage probabilities are reasonably close to the nominal 95%, with BOOT3 perhaps marginally the best. For this method the different factors considered have little effect on the coverage probabilities except for a few isolated cases like M1-F1-L5 and M1-F4-L3. Note that method BOOT1, yielding the shortest confidence interval lengths for distributions F1 and F2, gives very low coverage probabilities in these cases.

Remark. To complete the total picture it seems best to use BOOT3 in practice.

TABLE 7

Confidence interval lengths and coverage probabilities obtained by using the bootstrap methods.

(a) Confidence interval lengths

		TRU			BOOT1			BOOT2			BOOT3		
		L1	L3	L5	L1	L3	L5	L1	L3	L5	L1	L3	L5
F1	M1	89	92	97	97	98	97	115	116	113	110	109	105
	M3	95	95	92	89	90	90	110	110	108	107	107	104
	M5	88	83	86	93	85	82	111	104	100	109	102	98
F2	M1	108	99	100	119	117	118	130	131	127	125	123	118
	M3	97	98	111	106	107	107	123	123	121	121	119	116
	M5	104	104	101	110	115	117	128	132	131	124	127	126
F3	M1	108	115	111	133	132	134	134	136	131	128	128	122
	M3	109	114	117	118	120	121	128	130	127	128	129	125
	M5	108	144	144	121	130	133	135	146	147	133	142	144
F4	M1	112	176	112	201	176	192	137	143	131	136	134	124
	M3	111	103	112	156	155	165	133	136	132	136	131	124
	M5	173	338	315	167	250	274	158	237	248	186	228	240
F5	M1	295	301	297	705	506	557	308	324	294	299	294	270
	M3	289	313	318	637	425	461	304	313	300	309	301	282
	M5	342	*	*	826	*	*	733	889	791	*	*	919

*greater than 1000

(b) Coverage probabilities

		BOOT1			BOOT2			BOOT3		
		L1	L3	L5	L1	L3	L5	L1	L3	L5
F1	M1	91,6	92,4	91,2	96,8	95,6	95,2	94,8	95,6	92,4
	M3	92,8	90,4	91,6	96,8	94,8	94,4	96,4	94,0	92,8
	M5	93,6	92,0	90,8	98,0	97,6	96,0	97,6	96,4	96,4
F2	M1	92,4	94,4	97,2	96,0	96,4	97,2	94,4	96,0	95,2
	M3	92,0	93,6	94,8	96,0	95,2	95,6	95,6	94,4	94,8
	M5	92,0	91,2	92,0	96,4	97,6	97,2	95,2	96,8	96,4
F3	M1	95,2	94,8	96,4	95,6	96,0	96,8	96,0	95,6	94,8
	M3	91,2	92,8	94,4	95,6	95,6	94,8	94,4	94,8	94,0
	M5	94,8	92,0	88,0	96,4	96,0	92,8	95,6	94,8	92,4
F4	M1	97,6	97,2	98,0	98,4	98,8	95,6	97,2	98,0	96,0
	M3	95,6	96,4	96,8	97,2	97,6	94,8	97,6	96,4	95,2
	M5	93,6	93,2	93,2	98,4	94,0	94,8	96,4	95,6	94,8
F5	M1	98,4	96,8	98,4	97,2	96,4	96,0	96,0	95,2	94,4
	M3	98,4	97,6	98,4	97,2	94,8	95,6	95,6	94,4	94,0
	M5	94,8	90,4	90,4	96,4	93,2	92,0	95,2	94,4	93,2

7. CONCLUDING REMARKS

Taking all the results of the previous section into account one feels that, of the bootstrap methods, BOOT3 seems to be slightly better than the others in most of the cases, i.e. yielding the shortest confidence interval lengths and the right (95%) percentage coverage. Compared to this, method GROSS on the other hand outperforms BOOT3 in that it yields shorter confidence interval lengths and similar coverage probabilities. The only exceptions occur in the M5-F5 cases where methods GROSS yields too low percentage coverage. Similar conclusions can be drawn for the other sample sizes that were studied. However, since no suitable degree of freedom correction for compromise t^* values could be found, the method GROSS seems to be too complicated to use for practical purposes. BOOT3, although doing slightly worse than GROSS, is much easier to implement and is thus recommended for practical use.

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